



On the maximum penalized likelihood approach for proportional hazard models with right censored survival data



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ABSTRACT

This paper considers simultaneous estimation of the regression coefficients and baseline hazard in proportional hazard models using the maximum penalized likelihood (MPL) method where a penalty function is used to smooth the baseline hazard estimate. Although MPL methods exist to fit proportional hazard models, they suffer from the following deficiencies: (i) the positivity constraint on the baseline hazard estimate is either avoided or poorly treated leading to efficiency loss, (ii) the asymptotic properties of the MPL estimator are lacking, and (iii) simulation studies comparing the performance of MPL to that of the partial likelihood have not been conducted. In this paper we propose a new approach and aim to address these issues. We first model baseline hazard using basis functions, then estimate this approximate baseline hazard and the regression coefficients simultaneously. The penalty function included in the likelihood is quite general but typically assumes prior knowledge about the smoothness of the baseline hazard. A new iterative optimization algorithm, which combines Newton's method and a multiplicative iterative algorithm, is developed and its convergence properties studied. We show that if the smoothing parameter tends to zero sufficiently fast, the new estimator is consistent, asymptotically normal and retains full efficiency under independent censoring. A simulation study reveals that this method can be more efficient than the partial likelihood method, particularly for small to moderate samples. In addition, our simulation shows that the new estimator is substantially less biased under informative censoring.

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1. Introduction

The proportional hazard (or Cox's regression) model (e.g. Cox (1972) and Kalbfleisch and Prentice (2002, Chapter 4)) is one of the most widely used model for analyzing time to event (survival time) data. Regression coefficients of proportional hazard models, however, are usually estimated by maximizing the Cox's partial likelihood (PL) function (Cox, 1975), where the baseline hazard is not required when estimating the regression coefficients. The popularity of PL is mainly due to: (i) its ability to avoid estimating the baseline hazard function when the main focus is on estimating the regression coefficients, and (ii) asymptotic properties of the maximum partial likelihood estimate, such as in Tsiatis (1981) and Andersen and Gill (1982). When the baseline hazard is also of interest it can be estimated from the Breslow estimate (Breslow, 1972) or smoothed versions of it based on a kernel (Gray, 1990) or penalized likelihood smoothing (Anderson and Senthilselvan, 1980). The PL approach, however, is not always easy to extend to more general models or complex censoring schemes. For instance,

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such a partial likelihood is not available in a closed form for interval censored data (Sun, 2006) unless a PL-based estimating equation approximation (Datta et al., 2000) is used. Direct extensions of PL to additive or accelerated failure time models are not easy.

In this paper we adopt the MPL method to fit the Cox model. Earlier relevant references include Gray (1994), Joly et al. (1998) and Cai and Betensky (2003) where splines were used to approximate the baseline hazard for left or right censoring. Gray (1994) focused on testing hypotheses on covariate effects in the Cox model while others considered the estimation problem. Different penalty functions, such as roughness penalties, were considered.

However, existing methods suffer from some of the following deficiencies. Firstly, the way the positivity constraint on the baseline hazard is dealt with is suboptimal. It is either ignored (Gray, 1994) or ensured through “ad-hoc” techniques like the use of squares of splines coefficients (Joly et al., 1998), which potentially creates convergence issues (especially when some coefficients are zero). Others like Cai and Betensky (2003) prefer to model the log-baseline hazard as a linear spline, leading to difficulties in obtaining the cumulative baseline hazard in closed form, and also linear spline is inflexible. Secondly, the asymptotic properties are rarely discussed or rigorously presented. Thirdly, simulation studies comparing their performance to PL’s are lacking, casting doubt on the real advantages of these approaches. This paper attempts to address these issues for a new MPL proposal. Specifically the non-negativity constraint is ensured by directly restricting the coefficients of the basis functions, and we develop a new algorithm computing *simultaneously* the MPL estimates of the regression coefficients and baseline hazard. The asymptotic properties of the MPL estimator are derived when (a) the smoothing parameter converges to zero sufficiently fast, or (b) when it does not converge to zero. Its performance with respect to PL is assessed by simulations.

We develop our method in the context of proportional hazard model and right censoring in this paper. But it is, in principle, general enough to accommodate a large class of censoring schemes including interval censoring. Extensions to different survival models such as additive or accelerate failure time models are clearly possible but will be reported elsewhere.

This paper is organized as follows. In Section 2 we review the standard formulation of the Cox model and introduce the MPL approach. In Section 3 we propose a new alternating algorithm for computing the MPL estimates of the regression coefficients and baseline hazard and prove its convergence. The asymptotic properties of the MPL estimates are developed in Section 4, where consistency and asymptotic normality of the MPL estimator are obtained when the smoothing parameter is of order $o(n^{-1/2})$. In Section 5 a simulation study comparing the performance of PL and MPL under various levels of censoring is conducted. Under independent censoring MPL provides more efficient estimates than PL for small to moderate sample sizes, with the gain vanishing for large samples. Moreover, we find from the simulation that the MPL estimates are consistently less biased than the PL’s under dependent censoring, particularly for large censoring proportions and sample sizes. In some cases the gain in efficiency is substantial. The new approach is also illustrated using a critical care dataset in Section 5. Finally, concluding remarks are given in Section 6.

2. Fitting proportional hazard model by maximum penalized likelihood

2.1. Model and notation

For simplicity, we assume continuous survival times throughout this paper. We denote the independent survival times by $t_1, \dots, t_n$ (obtained from n individuals), where some of these observations are right censored. To simplify notations we use $\mathcal{C}$ and $\mathcal{O}$ to denote, respectively, the index set for censored and fully observed survival (or event) times; thus t_i with $i \in \mathcal{C}$ represents a right censored time, while t_i with $i \in \mathcal{O}$ presents an event time. Corresponding to each t_i , there are observations on a set of p covariates, and these observations are denoted by $x_{i1}, \dots, x_{ip}$. For each observation, the censoring time is assumed to be independent of the survival time. Let $h_i(t)$ be the hazard function of object i . The proportional hazard model specifies $h_i(t)$ according to

$$h_i(t) = h_0(t) \exp\{X_i\beta\}, \quad (1)$$

where $h_0(t)$ is the baseline hazard, $X_i = (x_{i1}, \dots, x_{ip})$ is the i th row of the design matrix X , and $\beta = (\beta_1, \dots, \beta_p)^T$ is the regression coefficient vector. Here the superscript T represents matrix transpose. We aim at estimating the coefficient vector β and the baseline hazard $h_0(t)$.

The traditional approach of estimating β , proposed by Cox (1972), is achieved by maximizing the PL function, given by

$$\mathcal{L}(\beta) = \prod_{i=1}^K \frac{\exp\{X_{(i)}\beta\}}{\sum_{r \in R(t_{(i)})} \exp\{X_r\beta\}}, \quad (2)$$

where $t_{(i)}$ is the i th ordered statistic of the uncensored t_i ’s, K is the total number of event times, $X_{(i)}$ is X_i corresponding to $t_{(i)}$ and $R(t_{(i)})$ is the set of individuals at risk at $t_{(i)}$. Note that this partial likelihood can be explained as the conditional probability of an individual who failed at $t_{(i)}$ given the risk set $R(t_{(i)})$ (e.g. Kalbfleisch and Prentice (2002, Chapter 4)). Once the estimate of β is available, the baseline hazard $h_0(t)$ can be obtained from, for example, the Breslow method; see Breslow (1972).

Although the PL estimate of β is asymptotically unbiased (Tsiatis (1981) and Andersen and Gill (1982)) in the case of independent censoring, it is known to be biased under dependent or informative censoring; see also the simulation results

in Section 5. We will demonstrate in this simulation that when the censoring is informative, MPL estimates of β developed under the independent censoring assumption have nevertheless a smaller bias and therefore a smaller MSE than their PL-based counterpart.

2.2. Maximum penalized likelihood (MPL)

For individual i , we let $H_i(t) = \int_0^t h_i(u)du$ be the cumulative hazard and $S_i(t)$ the survival function. The log-likelihood function from the observed survival/censoring times $t_1, \dots, t_n$ is (Kalbfleisch and Prentice, 2002, Chapter 6):

$$l(\beta, h_0) = - \sum_{i=1}^n H_i(t_i) + \sum_{i \in \mathcal{O}} \log h_i(t_i). \quad (3)$$

Substituting the proportional hazard (1) into log-likelihood (3) we have

$$l(\beta, h_0) = - \sum_{i=1}^n H_0(t_i) \exp\{X_i\beta\} + \sum_{i \in \mathcal{O}} (\log h_0(t_i) + X_i\beta), \quad (4)$$

where $H_0(t) = \int_0^t h_0(u)du$, the cumulative baseline hazard function.

Estimating h_0 by maximizing $l(\beta, h_0)$ is an ill-conditioned problem since h_0 is infinite dimensional while we only have finite number of observations. One way to circumvent this problem is to impose restrictions, such as a smoothness constraint, on h_0 by adding a penalty term to the log-likelihood to stabilize the estimation process and avoid large variances. Denote the penalty function by $J(h_0)$; we estimate β and $h_0(t)$ by maximizing the penalized log-likelihood function given by

$$\Phi(\beta, h_0) = l(\beta, h_0) - \lambda J(h_0), \quad (5)$$

where $\lambda \geq 0$ is the smoothing parameter used to balance two conflicting objectives: good fit to the data and compliance to the constraint imposed on $h_0(t)$. Note that the smoothing parameter can also depend on n but for simplicity the suffix is omitted. The penalty $J(h_0)$ in (5) restricts the estimation process of h_0 whereas β is free to vary. The main difficulty of this approach comes from the maximization of the penalized log-likelihood $\Phi(\beta, h_0)$ with respect to β and h_0 subject to the constraint $h_0 \geq 0$.

Manipulating the function h_0 directly can be difficult or even intractable sometimes. One common approach is to replace $h_0(t)$ by a function in a finite dimensional space. Assume $\psi_1, \dots, \psi_m$ form a basis of this finite dimensional space, then we set

$$h_0(t) = \sum_{u=1}^m \theta_u \psi_u(t). \quad (6)$$

Examples of basis functions include spline, kernel and indicator functions. We select a basis such that $\psi_u(t) \geq 0$ for all u , so that it suffices to restrict $\theta_u \geq 0$ for enforcing $h_0(t) \geq 0$. Let θ be the vector for all θ_u 's.

In this paper we develop our algorithm using this general basis function expression. But our simulation study in Section 5 will assume indicator basis functions. Indicator basis functions provide a piecewise constant (or step) baseline hazard function. More specifically, let $t_{(1)} = \min\{t_i : i = 1, \dots, n\}$ and $t_{(n)} = \max\{t_i : i = 1, \dots, n\}$. Assume sets $\{\mathcal{S}_1, \dots, \mathcal{S}_m\}$ form a partition to $\mathcal{D} = [t_{(1)}, t_{(n)}]$, i.e. $\cup_{u=1}^m \mathcal{S}_u = \mathcal{D}$ and $\mathcal{S}_u \cap \mathcal{S}_v = \emptyset$ if $u \neq v$, then the basis $\psi_u(t) = I(t \in \mathcal{S}_u)$, where I is an indicator function.

The problem can now be stated as maximizing the MPL objective function with respect to β and θ , with θ non-negative (i.e. all $\theta_u \geq 0$), namely

$$(\hat{\beta}, \hat{\theta}) = \underset{\beta, \theta}{\operatorname{argmax}} \{\Phi(\beta, \theta) = l(\beta, \theta) - \lambda J(\theta)\} \quad (7)$$

subject to $\theta \geq 0$, where

$$l(\beta, \theta) = - \sum_{u=1}^m \theta_u \sum_{i=1}^n \exp\{X_i\beta\} \psi_u(t_i) + \sum_{i \in \mathcal{O}} \left(\log \left(\sum_{u=1}^m \theta_u \psi_u(t) \right) + X_i\beta \right). \quad (8)$$

In (8), $\Psi_u(t_i) = \int_0^{t_i} \psi_u(v)dv$, which is in fact the corresponding basis of cumulative baseline hazard $H_0(t)$, i.e. $H_0(t) = \sum_{u=1}^m \theta_u \Psi_u(t)$. Different penalty functions are possible corresponding to approximation (6). For example, a roughness penalty, such as $\int h_0''(t)^2 dt$, can be written as $J(\theta) = \theta^T R \theta$, where R is an $m \times m$ matrix with elements $R_{ur} = \int \psi_u''(t) \psi_r''(t) dt$.

3. Algorithm

3.1. Principle

We propose an iterative algorithm to solve the optimization problem (7) under the constraint $\theta \geq 0$. In this algorithm, the two components β and θ are updated alternately in each iteration. Specifically, β is updated using the Newton–Raphson algorithm while θ is renewed with the multiplicative iterative (MI) algorithm of Ma (2010). Note that more traditional constrained optimization methods, such as those given in Luenberger (1984, Chapter 11), are not used for θ mainly because they are inefficient when the second derivative of Φ with respect to θ is poorly scaled (i.e. has a large condition number), which is common for the proportional hazard model. The MI algorithm only demands the first derivative, making it very easy to derive and implement.

The MI algorithm (Ma, 2010) was originally developed for penalized likelihood tomographic image reconstructions. These reconstruction problems are equivalent to constrained generalized linear models (GLMs) with huge dimensions (at least 128^2 non-negative regression parameters) and where the number of observations is usually smaller than the number of regression parameters. MI is an efficient non-negatively constrained algorithm which requires only the first derivative of the objective function. Its application can be extended to general constrained parameters estimation problems including hazard estimation.

The first derivatives of Φ with respect to β and θ are

$$\frac{\partial \Phi(\beta, \theta)}{\partial \beta_j} = - \sum_{i=1}^n H_0(t_i) \exp\{X_i \beta\} x_{ij} + \sum_{i \in \mathcal{O}} x_{ij}, \quad (9)$$

$$\frac{\partial \Phi(\beta, \theta)}{\partial \theta_u} = - \sum_{i=1}^n \psi_u(t_i) \exp\{X_i \beta\} + \sum_{i \in \mathcal{O}} \frac{\psi_u(t_i)}{h_0(t_i)} - \lambda \frac{\partial J(\theta)}{\partial \theta_u}, \quad (10)$$

for $j = 1, \dots, p$ and $u = 1, \dots, m$. The Karush–Kuhn–Tucker (KKT) necessary conditions for the constrained MPL estimation of β and θ are

$$\frac{\partial \Phi}{\partial \beta_j} = 0 \quad \text{for all } j, \quad (11)$$

$$\frac{\partial \Phi}{\partial \theta_u} = 0 \quad \text{if } \theta_u > 0 \quad \text{and} \quad \frac{\partial \Phi}{\partial \theta_u} < 0 \quad \text{if } \theta_u = 0, \quad \text{for } u = 1, \dots, m. \quad (12)$$

The algorithm proposed below can be used to solve Eqs. (11) and (12) iteratively. We call this algorithm the Newton multiplicative-iterative (Newton-MI) algorithm. Let $\beta^{(k)}$ and $\theta^{(k)}$ be respectively the estimate of β and θ at iteration k . Iteration $k + 1$ comprises two steps:

- Step 1: With θ set at $\theta^{(k)}$, $\beta^{(k+1)}$ is firstly obtained with one iteration of a modified Newton algorithm, where a line search step is included into this Newton iteration. This step is standard and ensures that the objective function $\Phi(\beta, \theta^{(k)})$ increases as a function of β when θ is maintained to its current value $\theta^{(k)}$.
- Step 2: Then with β fixed at $\beta^{(k+1)}$, $\theta^{(k+1)}$ is computed by running one iteration of the MI algorithm (Ma, 2010), followed by a line search to guarantee $\Phi(\beta^{(k+1)}, \theta)$ increases when moving from $\theta^{(k)}$ to $\theta^{(k+1)}$. This step also guarantees that each updated θ value respects the non-negativity constraint.

3.2. Details of the Newton-MI algorithm

Let A be a diagonal matrix with entries $H_0(t_1) \exp\{X_1 \beta\}, \dots, H_0(t_n) \exp\{X_n \beta\}$. We rewrite (9) as

$$\frac{\partial \Phi}{\partial \beta} = X^T (-A \mathbf{1}_n + \xi), \quad (13)$$

where $\mathbf{1}_n$ is an n -vector of 1s, and $\xi = (\xi_1, \dots, \xi_n)^T$ where $\xi_i = 1$ if $i \in \mathcal{O}$ and $\xi_i = 0$ for otherwise. From (13), the second derivative is

$$\frac{\partial^2 \Phi}{\partial \beta \partial \beta^T} = -X^T A X. \quad (14)$$

Thus one iteration of the Newton algorithm for solving (11) for β , emanating from $\beta^{(k)}$, gives

$$\beta^{(k+1)} = \beta^{(k)} + \omega^{(k)} (X^T A^{(k)} X)^{-1} X^T (-A^{(k)} \mathbf{1}_n + \xi), \quad (15)$$

where $A^{(k)}$ is A with β and θ replaced by $\beta^{(k)}$ and $\theta^{(k)}$ respectively and $\omega^{(k)} \leq 1$ is the line search step size. We can use, for example, Armijo's rule (Luenberger (1984, Chapter 7)) to determine the $\omega^{(k)}$'s. Inclusion of the line search assures that $\Phi(\beta^{(k+1)}, \theta^{(k)}) \geq \Phi(\beta^{(k)}, \theta^{(k)})$.

Parameter θ is updated using the MI algorithm of Ma (2010) such that each updated θ respects the positivity constraint. Firstly, (10) is rearranged such that both sides of the resulting equation are non-negative, namely,

$$\theta_u \left\{ \sum_{i=1}^n \psi_u(t_i) \exp\{X_i \beta\} + \lambda \left[\frac{\partial J}{\partial \theta_u} \right]^+ \right\} = \theta_u \left\{ \sum_{i \in \mathcal{O}} \frac{\psi_u(t_i)}{h_0(t_i)} - \lambda \left[\frac{\partial J}{\partial \theta_u} \right]^- \right\}, \quad (16)$$

where $[a]^+ = \max\{a, 0\}$ and $[a]^- = \min\{a, 0\}$ such that $a = [a]^+ + [a]^-$. This equation naturally suggests the following MI updating scheme

$$\theta_u^{(k+1/2)} = \theta_u^{(k)} \frac{\sum_{i \in \mathcal{O}} \frac{\psi_u(t_i)}{h_0^{(k)}(t_i)} - \lambda \left[\frac{\partial J(\theta^{(k)})}{\partial \theta_u} \right]^- + \varepsilon_u}{\sum_{i=1}^n \psi_u(t_i) \exp\{X_i \beta^{(k+1)}\} + \lambda \left[\frac{\partial J(\theta^{(k)})}{\partial \theta_u} \right]^+ + \varepsilon_u}, \quad (17)$$

where $h_0^{(k)}(t) = \sum_{u=1}^m \theta_u^{(k)} \psi_u(t)$ and $\partial J(\theta^{(k)})/\partial \theta_u = \partial J(\theta)/\partial \theta_u|_{\theta=\theta^{(k)}}$. In (17), $\varepsilon_u \geq 0$ are small constants and they are included to avoid zero denominator and to improve the convergence speed of Newton-MI; they do not alter the final solution of θ .

When the initial $\theta^{(0)} > 0$ then updates $\theta^{(k)}$ given by (17) are all non-negative. Moreover, if $\theta_u^{(k)} > 0$ then $\theta_u^{(k+1/2)} = 0$ only if $\sum_{i \in \mathcal{O}} \psi_u(t_i)/h_0^{(k)}(t_i) = 0$, $\partial J(\theta^{(k)})/\partial \theta_u \geq 0$ and $\varepsilon_u = 0$. Although it maintains that $\theta^{(k+1/2)} \geq 0$, iteration (17) may fail to increase $\Phi(\beta^{(k+1)}, \theta)$ when moving from $\theta^{(k)}$ to $\theta^{(k+1/2)}$, leading to possible divergence. To rectify this problem, a line search step is necessitated to give $\theta^{(k+1)}$ such that $\Phi(\beta^{(k+1)}, \theta^{(k+1)}) \geq \Phi(\beta^{(k+1)}, \theta^{(k)})$. Towards this, we first rewrite (17) as

$$\theta_u^{(k+1/2)} = \theta_u^{(k)} + s_u^{(k)} \frac{\partial \Phi(\beta^{(k+1)}, \theta^{(k)})}{\partial \theta_u}, \quad (18)$$

where $s_u^{(k)} = \theta_u^{(k)}/w_u^{(k)}$ with $w_u^{(k)}$ the denominator of the right hand side of (17). Clearly, when $\theta_u^{(k)} \neq 0$ we have $s_u^{(k)} > 0$. When $\theta_u^{(k)} = 0$ we set $s_u^{(k)} = 0$ only if $\partial \Phi(\beta^{(k+1)}, \theta^{(k)})/\partial \theta_u < 0$ (since $\theta_u^{(k)}$ satisfies the KKT condition in this case); otherwise, we set $s_u^{(k)} = c/w_u^{(k)}$, where c is a small constant such as 10^{-5} . Eq. (18) explains that $\theta_u^{(k+1/2)}$ emanates from $\theta_u^{(k)}$ in the gradient direction of θ with a non-negative step size $s_u^{(k)}$. In the context of line search, the search direction is $\theta^{(k+1/2)} - \theta^{(k)}$. Letting $\alpha^{(k)} > 0$ be the search step size, then $\theta^{(k+1)}$ is obtained according to

$$\theta^{(k+1)} = \theta^{(k)} + \alpha^{(k)} (\theta^{(k+1/2)} - \theta^{(k)}). \quad (19)$$

We only demand $\alpha^{(k)} \leq 1$ as such an $\alpha^{(k)}$ guarantees $\theta^{(k+1)} \geq 0$ when $\theta^{(k)} \geq 0$. The $\alpha^{(k)}$ value can be obtained by, for example, Armijo's rule.

3.3. Convergence properties

Due to the line search, the modified Newton algorithm in Step 1 guarantees the objective function $\Phi(\beta, \theta^{(k)})$ increases when β moves from $\beta^{(k)}$ to $\beta^{(k+1)}$. Similarly, the line search procedure in Step 2 ensures that $\Phi(\beta^{(k+1)}, \theta)$ increases when θ changes from $\theta^{(k)}$ to $\theta^{(k+1)}$. Thus after updating both parameter β and θ at iteration $k+1$, $\Phi(\beta^{(k+1)}, \theta^{(k+1)}) \geq \Phi(\beta^{(k+1)}, \theta^{(k)}) \geq \Phi(\beta^{(k)}, \theta^{(k)})$, where equalities hold only when iterations have converged. This property, together with the facts that β and θ updates are bounded and iteration mappings are continuous, leads to the convergence result of the Newton-MI algorithm. Results are stated in Theorem 1 and its proof is given in Appendix A.

Theorem 1. Consider the Newton-MI algorithm defined in Section 2. Assume $\beta \in R^p$, the p dimensional real space, and assume elements of θ are all non-negative, i.e. $\theta \geq 0$. Let M_1 and M_2 be respectively the iteration mappings for the modified Newton and the MI components of the Newton-MI algorithm, so that $\beta^{(k+1)} = M_1(\beta^{(k)}; \theta^{(k)})$ and $\theta^{(k+1)} = M_2(\theta^{(k)}; \beta^{(k+1)})$. Assume matrix A defined in Eq. (13) satisfies that $A^{1/2}X$ has full column rank for $\beta \in R^p$ and $\theta \geq 0$. With starting values $\theta^{(0)} > 0$ and $\beta^{(0)} \in R^p$, the Newton-MI algorithm produces a sequence $\{\beta^{(k)}, \theta^{(k)}\}$ converging to a solution satisfying the KKT conditions (11) and (12).

Proof. See Appendix A. $\square$

We give further comments on the Newton-MI algorithm.

1. From (14), $\partial^2 \Phi / \partial \beta \partial \beta^T$ is positive definite if $A^{1/2}X$ has full column rank for $\beta \in R^p$ and $\theta \geq 0$. Thus estimating β is relative easy and is less dependent on its starting value.
2. If we choose spline basis functions with roughness penalty the convergence is generally very fast.
3. If we use indicator basis functions with the bins selected such that each bin contains a fixed number of events, then this algorithm still converges fast in general.

4. Asymptotic properties of $\hat{\beta}$ and $\hat{\theta}$

In this section, we study the asymptotic properties of the MPL estimates $\hat{\beta}$ and $\hat{\theta}$. We restrict ourselves to asymptotics where the number of basis functions is fixed, though potentially large. To facilitate this presentation, let $\eta = (\beta^T, \theta^T)^T$ be the parameter vector to be estimated, whose length is $p + m$. Denote each survival or censoring observation as a bivariate random vector (t_i, ξ_i) , where ξ_i is a censoring indicator (i.e. $\xi_i = 1$ if t_i is an event time and $\xi_i = 0$ if it is a censoring time). Let $g_i(t_i, \xi_i; \eta)$ be its joint density function, conditional on the covariate X_i that is omitted for simplicity. Function $g_i(t_i, \xi_i; \eta)$ is given by

$$g_i(t_i, \xi_i; \eta) = \{f_i(t_i)S_i^c(t_i)\}^{\xi_i} \{f_i^c(t_i)S_i(t_i)\}^{(1-\xi_i)}, \quad (20)$$

where $f_i(t)$ is the density function of the event time, and $f_i^c(t_i)$ and $S_i^c(t_i)$ are respectively the density and survival functions of the censoring time. Hence the log-likelihood (3) can be rewritten as

$$l(\eta) = \sum_{i=1}^n l_i(\eta) = \sum_{i=1}^n \log g_i(t_i, \xi_i; \eta). \quad (21)$$

We rewrite the objective function in (7) as

$$\Phi(\eta) = \sum_{i=1}^n (l_i(\eta) - \mu_n J(\eta)), \quad (22)$$

where $J(\eta) = J(\theta)$ and $\mu_n = \lambda/n$.

4.1. Asymptotics when $\mu_n \rightarrow 0$

The following conditions are needed to develop the asymptotic properties stated in Theorem 2.

1. The triples (t_i, ξ_i, X_i) , $1 \leq i \leq n$, are independently and identically distributed, and the distribution of X_i is independent of η .
2. The censoring time is independent of the true possibly unobserved survival time conditionally on the covariates.
3. $\lim_{n \rightarrow \infty} n^{-1} \sum_{i=1}^n l_i(\eta)$ exists and has a unique maximum at $\eta = \eta_0$, an interior point of the parameter space Γ which is assumed compact.
4. μ_n is $o(n^{-1/2})$.
5. There exists a measurable function $b(t, \xi, x)$ with $E(b(t, \xi, x)) < \infty$ satisfying $|\log g_i(t_i, \xi_i; \eta)| \leq b(t_i, \xi_i, X_i)$ for all η in the parameter space.
6. $l(\eta)$ is twice differentiable in a neighborhood of η_0 and the Fisher information matrix

$$\mathcal{I}(\eta_0) = \lim_{n \rightarrow \infty} n^{-1} \sum_{i=1}^n \frac{\partial l_i(\eta)}{\partial \eta} \frac{\partial l_i(\eta)}{\partial \eta^T} \quad (23)$$

exists and is positive definite. Moreover the matrices

$$n^{-1} \sum_{i=1}^n \frac{\partial l_i(\eta)}{\partial \eta} \frac{\partial l_i(\eta)}{\partial \eta^T} \quad \text{and} \quad n^{-1} \sum_{i=1}^n \frac{\partial^2 l_i(\eta)}{\partial \eta \partial \eta^T} \quad (24)$$

converge uniformly in η in a neighborhood of η_0 .

7. For the penalty function, $J(\eta)$ is positive and both $\partial J(\eta)/\partial \eta$ and $\partial^2 J(\eta)/\partial \eta \partial \eta^T$ exist. Also, $\partial^2 J(\eta)/\partial \eta \partial \eta^T$ is bounded in a neighborhood of η_0 .

Conditions (1)–(3) are standard conditions used in maximum likelihood theory with censored data. Condition (4) appeared first in Yu and Ruppert (2002) in their asymptotic results for a penalized splines estimator in partially linear single-index models. It was also mentioned as a hint in Kauermann (2005). It is required to prove the asymptotic normality of the MPL although only a weaker condition (i.e. $\mu_n = o(1)$) is required to prove consistency. Conditions (5) and (6) are technical conditions needed to prove consistency and asymptotic normality—see Appendix B for details. Condition (5) replaces the stringent Assumption 2 in Yu and Ruppert (2002, page 1045), while Condition (6) is similar to their Assumption 4. Condition (7) is needed to control the penalty term in the proof of asymptotic normality and this condition is typically satisfied for quadratic penalties that we are mainly interested in.

Theorem 2. Assume that conditions (1)–(7) hold then the MPL estimator $\hat{\eta}$ is strongly consistent for η_0 and the distribution of $\sqrt{n}(\hat{\eta} - \eta_0)$ converges, when $n \rightarrow \infty$, to the multivariate normal distribution $N(0_{(p+m) \times 1}, \mathcal{I}(\eta_0)^{-1})$.

Proof. See Appendix B. $\square$

Theorem 2 shows that when μ_n goes to zero sufficiently fast the MPL estimate $\hat{\beta}$ is asymptotically unbiased and efficient (i.e. it has the same asymptotic variance as the corresponding maximum likelihood obtained when $\mu_n = 0$). Its performance relative to PL in small to moderate samples is unknown but possible gains in efficiency are expected due to the fact that it is based on a full likelihood. This will be further confirmed by simulations in Section 5.1. Note that

$$\frac{\partial^2 l(\eta)}{\partial \eta \partial \eta^T} = \begin{pmatrix} \frac{\partial^2 l(\beta, \theta)}{\partial \beta \partial \beta^T} & \frac{\partial^2 l(\beta, \theta)}{\partial \beta \partial \theta^T} \\ \frac{\partial^2 l(\beta, \theta)}{\partial \theta \partial \beta^T} & \frac{\partial^2 l(\beta, \theta)}{\partial \theta \partial \theta^T} \end{pmatrix}, \quad (25)$$

where

$$\frac{\partial^2 l}{\partial \beta \partial \beta^T} = -X^T A X, \quad \frac{\partial^2 l}{\partial \theta \partial \theta^T} = -P^T D_1 P, \quad \frac{\partial^2 l}{\partial \beta \partial \theta^T} = -X^T D_2 G.$$

Here matrix $P = (\psi_u(t_i))_{n \times m}$, matrix $D_1 = \text{diag}(\xi_1/h_0(t_1)^2, \dots, \xi_n/h_0(t_n)^2)$, matrix $D_2 = \text{diag}(\exp\{X_1\beta\}, \dots, \exp\{X_n\beta\})$ and matrix $G = (\Psi_u(t_i))_{n \times m}$. These expressions provide an easy way to estimate the Fisher information matrix and the asymptotic variances.

In the case of $\mu_n = 0$ (i.e. ML) and $h_0(t)$ is discretized, estimates of β and $h_0(t)$ can be obtained using Poisson regression with a log link function; see [Whitehead \(1980\)](#) and [Frome \(1983\)](#).

4.2. Asymptotics and inference when μ_n is fixed

The asymptotic distribution of the MPL estimate given in [Theorem 2](#) does not depend on μ_n as μ_n goes to zero sufficiently fast. We now give asymptotic results when $\mu_n = \mu$, a fixed value independent on n . In that case, the MPL estimate $\hat{\eta}$ solves the first-order equation

$$\sum_{i=1}^n \left(\frac{\partial}{\partial \eta} l_i(\eta) - \mu \frac{\partial J(\eta)}{\partial \eta} \right) = 0 \quad (26)$$

with respect to η . This can be rewritten as an estimating equation $\sum_{i=1}^n \kappa_i(\eta; \mu) = 0$ where

$$\kappa_i(\eta; \mu) = \frac{\partial}{\partial \eta} l_i(\eta) - \mu \frac{\partial J(\eta)}{\partial \eta}. \quad (27)$$

Assume that η^* satisfies $\sum_{i=1}^n E(\kappa_i(\eta^*; \mu)) = 0$ then, since μ is fixed, $\hat{\eta}$ is a consistent estimator of η^* given by an unbiased estimating equation. Both $\hat{\eta}$ and η^* are obviously functions of μ but this dependence is omitted for simplicity. The covariance matrix of $\hat{\eta}$ is given by a sandwich formula (see [Huber \(1967\)](#), [Carroll and Stefanski \(1995, Appendix 3\)](#) and [Yu and Ruppert \(2002\)](#) for its use in the penalized likelihood estimation setting). Define

$$\hat{M}(\eta) = n^{-1} \sum_{i=1}^n \frac{\partial}{\partial \eta} \kappa_i(\eta) \quad \text{and} \quad \hat{Q}(\eta) = n^{-1} \sum_{i=1}^n \kappa_i(\eta) \kappa_i(\eta)^T. \quad (28)$$

If $\hat{M}(\eta)$ and $\hat{Q}(\eta)$ have limits M and Q as $n \rightarrow \infty$, under similar assumptions to Conditions 1–7 and by using the same kind of argument used in [Theorem 2](#), one can show that $\hat{\eta}$ is consistent for η^* and that $\sqrt{n}(\hat{\eta} - \eta^*)$ is asymptotically normal $N(0, V(\eta^*))$ with $V(\eta^*) = M(\eta^*)^{-1} Q(\eta^*) M(\eta^*)^{-1}$. Straightforward calculations show that

$$V(\eta^*) = \left[\mathcal{I}(\eta^*) + \mu \frac{\partial^2 J(\eta^*)}{\partial \eta \partial \eta^T} \right]^{-1} Q(\eta^*) \left[\mathcal{I}(\eta^*) + \mu \frac{\partial^2 J(\eta^*)}{\partial \eta \partial \eta^T} \right]^{-1}, \quad (29)$$

where

$$\frac{\partial^2 J(\eta)}{\partial \eta \partial \eta^T} = \begin{pmatrix} 0 & 0 \\ 0 & \frac{\partial^2 J(\theta)}{\partial \theta \partial \theta^T} \end{pmatrix}.$$

This sandwich formula is valid for a fixed but possibly large dimension of η and is similar in structure to [Gray \(1994, Eq. \(A.3\)\)](#). In practice a consistent estimate of $V(\eta^*)$ is obtained by replacing $\mathcal{I}(\eta^*)$ and $Q(\eta^*)$ by their empirical versions and plugging-in $\hat{\eta}$ for η^* in (29). The assumption $\mu_n = \mu$ can be slightly relaxed, i.e. when $\mu_n \rightarrow \mu \neq 0$ the same asymptotic result still holds.

In general, the MPL estimate is biased for η_0 unless we impose strong conditions on μ_n . Condition (4) in [Theorem 2](#) seems to be critical. [Yu and Ruppert \(2002\)](#) investigated this in the single index model with only one covariate. They showed that for quadratic penalties, when $\mu_n = Cn^{-1/2}$ ($C > 0$), the asymptotic normality of the MPL estimates of spline coefficients is preserved but the mean is shifted away from its intended target. In addition the asymptotic bias of $\sqrt{n}(\hat{\eta} - \eta_0)$ is of order

$O(\mu_n)$ so that if $\mu_n \rightarrow 0$ slower than $n^{-1/2}$ then the bias dominates. The bias depends on how close η^* is to η_0 , but under general circumstances this quantity is difficult to quantify although attempts in this direction were made by Green (1987) and Fessler (1996) but they do not deal with censored data. Bias-correction techniques based on indirect inference (Moustaki and Victoria-Feser, 2006) could in principle be used but would be cumbersome and probably inefficient.

The previous derivations of the asymptotic variance of $\hat{\eta}$ assume that $\eta = (\beta^T, \theta^T)^T$ is an interior point of the parameter space. It is, however, possible that θ elements take zero values. It is difficult to develop the asymptotic theory for $\hat{\beta}$ and $\hat{\theta}$ when some θ elements lies on the boundary. In this case, bootstrapping can provide approximate mean and variance, but it requires a large amount of repeated estimations of both parameters, which can be computationally intensive. Our simulation in Section 5 however suggests that the asymptotic variance formula (29) is generally accurate even when some θ estimates are zero.

5. Results

This section reports the results from a simulation study and from a real data application. In the simulation study our aims are threefold: (i) demonstrate that the proposed method of MPL estimation of β and θ works well, (ii) assess the efficiency of the MPL estimates and compare them to their PL counterpart under independent and dependent censoring schemes, and (iii) evaluate the asymptotic variance of the MPL estimate of β . All the results in this section are obtained using MATLAB. Although only one simulation is reported in this section, other simulations had also been conducted and similar results were observed.

The basis functions used in this section are indicator functions, leading to a piecewise constant (or discretization) approximation to $h_0(t)$. To simplify notations we denote the discretized $h_0(t)$ by vector $(\theta_1, \dots, \theta_m)^T$, where m is the number of bins for the discretization. In this simulation we choose the equal bin event count, where the common event count of each bin is denoted by n_0 . This choice is more efficient than equal bin width as it better manipulates the information contained in the data. In the simulation scenarios below we report n_0 in each example.

Corresponding to discretization, any derivative operation in a penalty function should be replaced by difference. The square of the first order difference penalty, namely

$$J(\theta) = \sum_{i=2}^m (\theta_i - \theta_{i-1})^2, \quad (30)$$

is therefore used for all the examples in this section. This penalty usually gives fast convergence of the Newton-MI algorithm and is sufficient for neighboring θ values to be similar. The choice of the penalty function is a matter of personal preference but may affect the small sample performance of MPL.

5.1. A simulation with independent censoring

In this example, we simulate survival times y_i from the Weibull distribution with hazard

$$h_i(t) = 3t^2 \exp\{x_{i1} + 0.1x_{i2} - 0.5x_{i3}\}. \quad (31)$$

For each y_i a censoring time c_i is generated independently of y_i from the exponential distribution $\text{Exp}(\mu_c)$. The mean μ_c is calibrated to achieve, on average, a pre-specified censoring level π_c . The observed data are $\{(t_i, \xi_i), i = 1, \dots, n\}$, where $t_i = \min\{y_i, c_i\}$ and $\xi_i = I(y_i < c_i)$, where I is the indicator function. Note that c_i in this example are generated independently of t_i .

Model (31) is a proportional hazard model with baseline hazard $h_0(t) = 3t^2$ and regression coefficients $\beta_1 = 1, \beta_2 = 0.1$ and $\beta_3 = -0.5$. Values for covariates x_1, x_2 and x_3 are generated from uniform distributions: $x_{i1} \sim \text{unif}(0, 1)$, $x_{i2} \sim \text{unif}(0, 5)$ and $x_{i3} \sim \text{unif}(0, 7)$.

To have a clear picture on how each estimation method is affected by the sample size n and censoring proportion, we choose $n = 100, 500$, and 5000 corresponding respectively to small, intermediate and large sample sizes, with approximate censoring proportions π_c of 20%, 50% and 80% for each value of n . Regression coefficients β_1, β_2 and β_3 are estimated by Cox's PL and our MPL.

As mentioned above, the “equal bin event count” strategy is used to define the indicator functions. Preliminary tests showed that in general the β estimates are not very sensitive to the choice of the common bin event count n_0 as long as it is not too large and the smoothing value is selected appropriately. In practice, we set $n_0 = 1$ for $n = 100$, $n_0 = 3$ for $n = 500$ and $n_0 = 10$ for $n = 5000$, decreasing the ratio n_0/n as more data are available. Convergence of the Newton-MI algorithm is achieved if the difference between updates of β and θ from two consecutive iterations is less than 10^{-5} . In the tables below we present $\lambda = \gamma/(1 - \gamma)$, so that the tuning parameter γ satisfies $\gamma \in [0, 1)$. The γ values given in Table 1 are chosen experimentally and are not particularly fine-tuned. They are nevertheless small enough to guarantee consistency for the regression parameter (see Theorem 2). Note that small samples generally need lighter smoothing while large samples require heavier smoothing; this is because fewer bins are used for small samples than for large samples.

For each combination of n and π_c , 500 repeated samples are generated and thus 500 estimates of β and θ are obtained. Performance of different β estimates are assessed by examining their biases and standard deviations (stds) or mean squared errors (MSEs), all obtained from the Monte Carlo simulation.

Table 1

Regression coefficients comparison using simulated samples with sizes 100, 500, and 5000. For each sample size there are approximately 20%, 50% and 80% independent censoring. Biases and stds are computed from 500 repeated samples and the asymptotic stds are obtained from the last of the 500 samples.

			$n = 100$			$n = 500$			$n = 5000$		
π_c			20%	50%	80%	20%	50%	80%	20%	50%	80%
n_0			1	1	1	3	3	3	10	10	10
γ			0.01	0.001	0.0001	0.3	0.05	0.01	0.4	0.6	0.7
$\beta_1 = 1$	PL	bias	0.024	0.057	0.069	0.012	0.010	0.057	0.004	−0.002	0.007
		std	0.423	0.589	1.036	0.185	0.233	0.363	0.059	0.078	0.112
		MSE	(0.1791)	(0.3507)	(1.0781)	(0.0342)	(0.0543)	(0.1350)	(0.0035)	(0.0061)	(0.0125)
	MPL	bias	−0.023	0.014	0.026	−0.005	−0.018	0.019	−0.005	−0.010	−0.003
		std	0.408	0.568	1.008	0.180	0.227	0.313	0.058	0.077	0.111
		astd	0.432	0.674	1.078	0.198	0.233	0.391	0.057	0.073	0.113
		MSE	(0.1666)	(0.3222)	(1.0169)	(0.0325)	(0.0516)	(0.1251)	(0.0034)	(0.0060)	(0.0122)
	PL	bias	0.004	0.003	0.019	0.0006	−0.003	−0.002	0.0001	−0.0005	0.0009
		std	0.085	0.117	0.203	0.035	0.045	0.078	0.011	0.014	0.023
		MSE	(0.0073)	(0.0137)	(0.0416)	(0.0012)	(0.0020)	(0.0061)	(0.0001)	(0.0002)	(0.0005)
$\beta_2 = 0.1$	MPL	bias	−0.001	−0.0009	0.014	−0.001	−0.006	−0.005	−0.0008	−0.001	0.0000
		std	0.082	0.113	0.196	0.034	0.044	0.076	0.010	0.014	0.022
		astd	0.082	0.119	0.228	0.038	0.045	0.072	0.011	0.014	0.023
		MSE	(0.0067)	(0.0127)	(0.0387)	(0.0012)	(0.0020)	(0.0057)	(0.0001)	(0.0002)	(0.0005)
	PL	bias	−0.020	−0.023	−0.064	−0.003	−0.004	−0.008	−0.0007	−0.0002	−0.0008
		std	0.080	0.103	0.191	0.034	0.043	0.068	0.010	0.013	0.021
		MSE	(0.0068)	(0.0111)	(0.0405)	(0.0011)	(0.0019)	(0.0047)	(0.0001)	(0.0002)	(0.0004)
	MPL	bias	0.001	−0.002	−0.042	0.002	0.008	0.009	0.003	0.002	0.004
		std	0.077	0.099	0.188	0.033	0.042	0.066	0.010	0.013	0.020
		astd	0.073	0.105	0.194	0.035	0.039	0.063	0.010	0.013	0.020
		MSE	(0.0059)	(0.0097)	(0.0371)	(0.0012)	(0.0018)	(0.0044)	(0.0001)	(0.0002)	(0.0004)

Results for β are summarized in Table 1, where the asymptotic std (astd) values are computed using the last of the 500 repeated samples and formula (29). Comparing astd against std in Table 1 instantiates that the astd formula (29) generally gives accurate variance approximation, particularly when the sample size is large or the censoring proportion is small. We observe, for both PL and MPL, that bias and variance increase with the censoring proportion but decrease with the sample size. In this example, MPL is at least as efficient as PL. In fact, MPL gives slightly better bias and std, and thus better MSE, for small to moderate sample sizes with large censoring proportions.

Since the MSE values are all small, we also computed and plotted their relative MSEs (e.g. $\text{MSE}(\hat{\beta}_1^{\text{MPL}})/\text{MSE}(\hat{\beta}_1^{\text{PL}})$ for β_1) in Fig. 1, which also includes in panel (d) the plot of overall relative MSEs (defined as the sum of MPL MSEs over the sum of PL MSEs). A ratio lower than 1 favors MPL. The overall MSE plot indicates a similar performance between the MPL and PL estimates for $n = 5000$. In general, MPL can produce better MSE than PL for a small sample. For example, MPL achieves about 13% MSE reduction over PL when estimating β_3 for $n = 100$ and $\pi_c = 20\%$.

5.2. Simulation results with informative censoring

We also study the performance of PL and MPL under dependent censoring although none of them is particularly built to deal with this situation. Here the objective is to see how sensitive the PL and MPL estimates are when informative censoring is present but ignored (as sometimes dependent censoring cannot be easily identified).

The model used in the previous section to generate survival data is also used here. Again random survival times $y_1, \dots, y_n$ are generated from the Weibull distribution with the same hazard function as (31). The censoring status of each y_i is however generated differently. First for a given censoring proportion π_c we decide whether an observation y_i is to be censored. If so, the corresponding censoring time t_i is generated from the $\text{unif}(0, y_i)$ distribution and the corresponding censoring indicator ξ_i is set to 0; otherwise, $t_i = y_i$ and ξ_i is set to 1. Clearly, censoring times generated in this way are dependent on the true survival times, leading to informative censoring. For consistency, the sample sizes, censoring proportions, and penalty function remain unchanged from the previous example. The “equal bin event count” option for MPL, where $n_0 = 1$ for $n = 100$, $n_0 = 3$ for $n = 500$ and $n_0 = 20$ for $n = 5000$, is used. The smoothing values are selected again experimentally; they are $\gamma = 0.8$ for $n = 100$ and $\gamma = 0.9$ for both $n = 500$ and $n = 5000$. In general larger smoothing parameters than for the independent censoring case need to be used. This is required to give more accurate θ estimates probably leading to less biased β estimates.

Table 2 contains biases and stds for the β estimates computed from 500 repeated samples. Results show that both the MPL and PL estimates are biased, but the bias of PL is much larger, especially when the censoring proportion is large. Also, for a fixed sample size, bias and variance tend to increase with the censoring proportion for both methods but MPL resists better. Fig. 2 exhibits the plots of relative MSEs. Clearly, MPL outperforms PL in this example, producing substantially lower

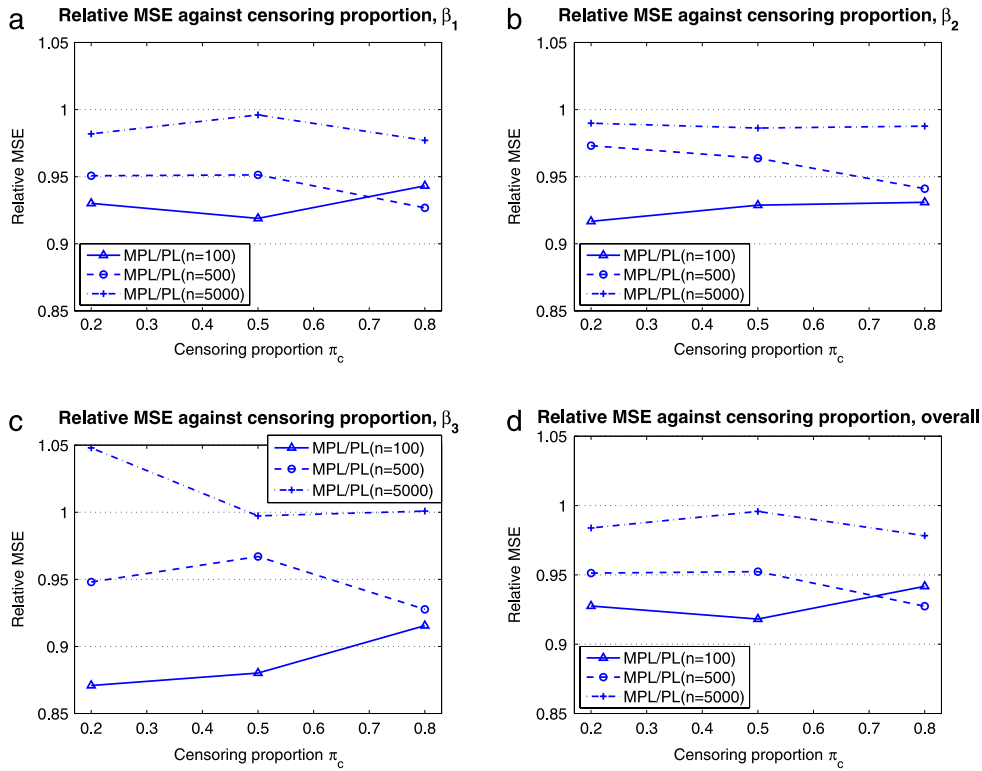


Fig. 1. Plots of the relative MSE for β_1 , β_2 , β_3 (panels (a), (b) and (c) respectively) and plot of the overall relative MSE (panel (d)), where censoring is independent.

Table 2

Regression coefficients comparison using simulated samples with sizes 100, 500, and 5000. For each sample size there are about 20%, 50% and 80% dependent censoring. Bias and std are computed from 500 repeated samples and the asymptotic std (astd) are obtained from the last of the 500 samples.

			$n = 100$			$n = 500$			$n = 5000$		
π_c			20%	50%	80%	20%	50%	80%	20%	50%	80%
n_0			1	1	1	3	3	3	20	20	20
γ			0.8	0.8	0.8	0.9	0.9	0.9	0.9	0.9	0.9
$\beta_1 = 1$	PL	bias	0.042	0.136	0.298	0.036	0.110	0.237	0.024	0.092	0.228
		std	0.449	0.600	0.990	0.174	0.230	0.344	0.055	0.072	0.116
		MSE	(0.2035)	(0.3782)	(1.0677)	(0.0316)	(0.0651)	(0.1744)	(0.0036)	(0.0135)	(0.0655)
	MPL	bias	-0.028	-0.003	0.033	-0.016	0.024	0.063	0.006	0.053	0.141
		std	0.416	0.545	0.994	0.167	0.216	0.321	0.054	0.070	0.111
		astd	0.414	0.585	0.954	0.181	0.242	0.368	0.057	0.072	0.114
		MSE	(0.1742)	(0.2967)	(0.9899)	(0.0280)	(0.0470)	(0.1073)	(0.0030)	(0.0077)	(0.0324)
		bias	0.003	0.012	0.025	0.005	0.009	0.024	0.003	0.008	0.022
		std	0.084	0.123	0.196	0.036	0.045	0.073	0.011	0.013	0.022
		MSE	(0.0070)	(0.0153)	(0.0389)	(0.0013)	(0.0021)	(0.0059)	(0.0001)	(0.0002)	(0.0010)
$\beta_2 = 0.1$	PL	bias	-0.004	0.0009	-0.0009	-0.0003	0.001	0.006	0.0008	0.004	0.013
		std	0.079	0.112	0.191	0.035	0.042	0.070	0.011	0.013	0.021
		astd	0.078	0.101	0.181	0.037	0.047	0.069	0.011	0.014	0.022
	MPL	bias	0.078	0.101	0.181	0.037	0.047	0.069	0.011	0.014	0.022
		std	0.069	0.089	0.164	0.031	0.036	0.058	0.010	0.012	0.019
		astd	0.068	0.097	0.170	0.032	0.042	0.065	0.010	0.013	0.021
		MSE	(0.0047)	(0.0082)	(0.0285)	(0.0009)	(0.0014)	(0.0048)	(0.0001)	(0.0010)	(0.0059)
		bias	-0.030	-0.072	-0.173	-0.017	-0.047	-0.116	-0.013	-0.045	-0.110
		std	0.083	0.109	0.186	0.035	0.042	0.071	0.010	0.012	0.021
		MSE	(0.0078)	(0.0169)	(0.0648)	(0.0015)	(0.0040)	(0.0185)	(0.0003)	(0.0021)	(0.0126)
$\beta_3 = -0.5$	PL	bias	-0.0006	-0.016	-0.040	0.003	-0.012	-0.038	-0.006	-0.030	-0.074
		std	0.069	0.089	0.164	0.031	0.036	0.058	0.010	0.012	0.019
		astd	0.068	0.097	0.170	0.032	0.042	0.065	0.010	0.013	0.021
	MPL	bias	0.069	0.089	0.164	0.031	0.036	0.058	0.010	0.012	0.019
		std	0.068	0.097	0.170	0.032	0.042	0.065	0.010	0.013	0.021
		MSE	(0.0047)	(0.0082)	(0.0285)	(0.0009)	(0.0014)	(0.0048)	(0.0001)	(0.0010)	(0.0059)
		bias	-0.030	-0.072	-0.173	-0.017	-0.047	-0.116	-0.013	-0.045	-0.110
		std	0.083	0.109	0.186	0.035	0.042	0.071	0.010	0.012	0.021
		MSE	(0.0078)	(0.0169)	(0.0648)	(0.0015)	(0.0040)	(0.0185)	(0.0003)	(0.0021)	(0.0126)

MSEs in all the cases. A MSE reduction ranging from 10% to 75% is observed for MPL. This simulation illustrates the extreme sensitivity of the PL approach to the independent censoring assumption, especially under heavy censoring. This could be

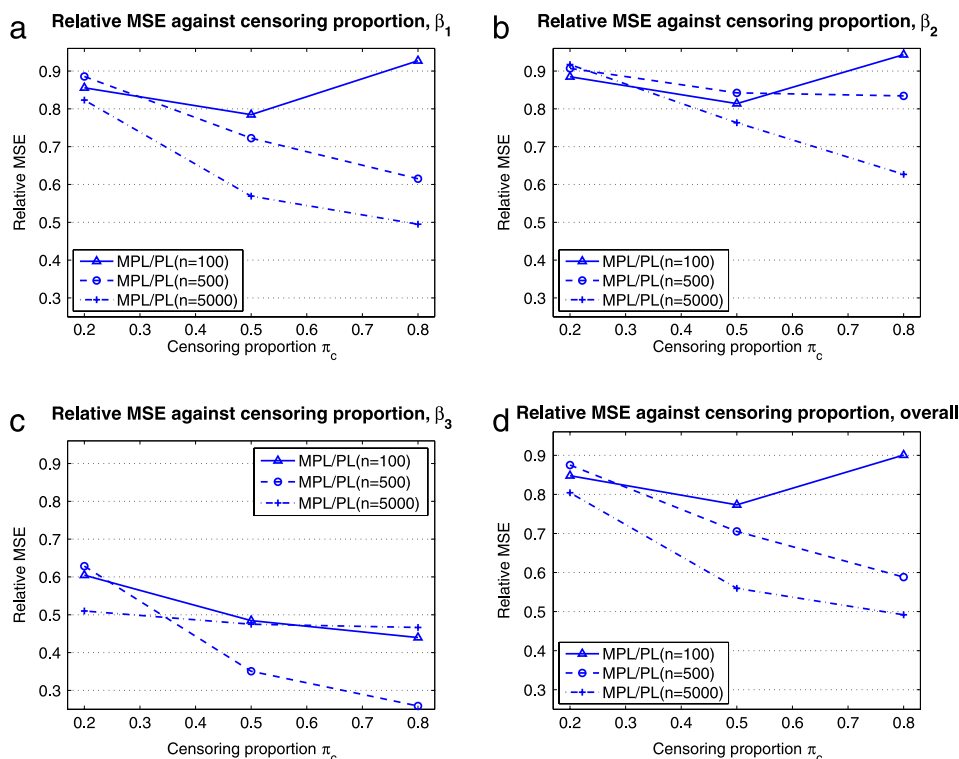


Fig. 2. Plots of the relative MSE for β_1 , β_2 , β_3 (panels (a), (b) and (c) respectively) and plot of the overall relative MSE (panel (d)), where censoring is informative.

due to the fact that PL is rank-based and thus incorrect risk sets are used in the PL equation when informative censoring occurs. In contrast, the (penalized) likelihood function does not rely on ranks, leading to substantially smaller bias in the β estimates. MPL provides further improvement over ML (smoothing parameter set to 0 in (7)) possibly because the penalty term helps to estimate the baseline hazard more accurately, which might be useful to correct the bias of the regression coefficient estimates. Evidence of this was observed in different simulations with and without the penalty term (results not shown) for different baseline hazards.

Similar results (not shown in this paper) were also observed from other dependent censoring schemes, some of them displaying less dramatic biases than those reported in this section. We would like to add that appropriate methods with “built-in” ability to cope with dependent censoring, such as copulas, should preferably be used and would be more efficient in these situations.

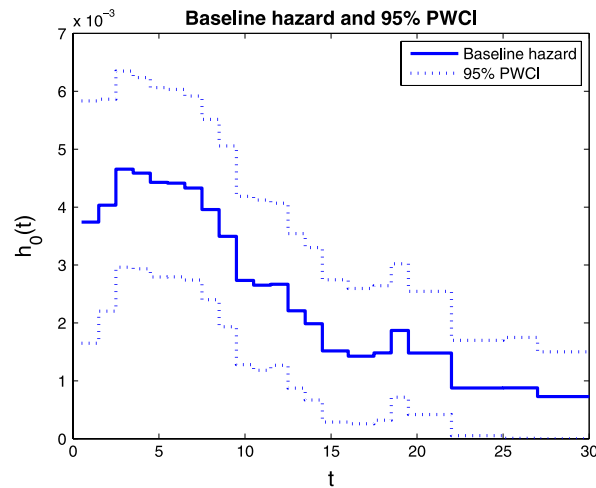
5.3. Traumatic brain injury study

In this section we apply the Newton-MI algorithm to fit proportional hazard models using the data from the traumatic brain injury study (The SAFE Study Investigators, 2007). This study evaluated two fluid resuscitation methods, the saline versus albumin methods, in patients with traumatic brain injury (TBI). A patient was defined to have traumatic brain injury if the patient had a history of trauma, evidence of head trauma on a CT scan, and a score of ≤ 13 on the Glasgow Coma Scale (GCS). This study was a collaboration of the Australia and New Zealand Intensive Care Society Clinical Trials Group, the Australia Red Cross Blood Service and the George Institute for International Health. Initially, 460 TBI patients of traumatic brain injury were included in this study, of whom 50.2% received albumin and 49.8% saline. After removing those patients with missing observations and other data cleaning, 412 patients were kept in the sample. The main endpoint is survival time in days (up to day 28) but patients were followed up for 24 months. Censoring is important in these data but most of it occurs at the end-of-study. The relevant covariates considered here are: AGE (in years), APACHE II (acute physiology and chronic health evaluation II score measuring injury severity), GCS (Glasgow coma scale), MAP (mean arterial pressure), TRT (indicator variable for saline or albumin treatments; 0 for albumin and 1 for saline), DIFF (swelling indicator; 1 for yes and 0 for no) and MLI (mass lesion indicator; 1 for yes and 0 for no). A more detailed description of these covariates, the study design and other considerations can be found in The SAFE Study Investigators (2007).

We first fitted a proportional hazard model, which includes all the covariates specified above. We used both Cox’s PL and our MPL to fit the model, and for MPL we used the indicator basis functions where the bins were determined by the “equal bin event count” strategy with 2 events in each bin. The smoothing value used was $\gamma = 1 - 10^{-5}$. Again, this smoothing

Table 3Regression coefficients estimates given by Cox's PL and MPL. Values in brackets are the corresponding p -values.

	AGE	APACHE II	GCS	MAP	TRT	DIFF	MLI
PL	0.022	0.701	−0.171	−0.001	−0.631	0.619	0.389
std	0.006	0.170	0.038	0.007	0.196	0.265	0.215
p -value	(0.0001)	(0.0000)	(0.0000)	(0.8901)	(0.0013)	(0.0195)	(0.0705)
MPL	0.023	0.731	−0.179	−0.001	−0.665	0.647	0.414
std	0.006	0.175	0.039	0.008	0.203	0.272	0.221
p -value	(0.0001)	(0.0000)	(0.0000)	(0.8979)	(0.0010)	(0.0175)	(0.061)

**Fig. 3.** Plots of the baseline hazard and its 95% pointwise confidence interval (PWCI) for the traumatic brain injury study.**Table 4**

Regression coefficients of proportional hazard models corresponding to albumin and saline treatments.

	AGE	APACHE II	GCS	MAP	DIFF	MLI
Albumin	0.020	0.775	−0.197	−0.006	0.752	0.597
std	0.007	0.222	0.048	0.010	0.332	0.277
p -values	(0.0066)	(0.0005)	(0.0000)	(0.5390)	(0.0235)	(0.0309)
Saline	0.028	0.662	−0.125	0.007	0.445	0.062
std	0.009	0.292	0.066	0.012	0.467	0.358
p -values	(0.0029)	(0.0230)	(0.0608)	(0.5849)	(0.3403)	(0.8634)

values was chosen to give a sufficiently smooth baseline hazard estimate and no effort was made to optimize this smoothing value. An “optimal” smoothing value can be found using, for example, the cross-validating procedure described in O’Sullivan (1988). Other criteria involving a grid search over several values of λ and the maximization of some sort of criteria like (cross-validated) AIC or BIC can also be used (see Perperoglou, 2013 for a recent reference). Results are summarized in Table 3. Clearly, Cox’s PL and our MPL give very similar regression coefficient estimates, indicating that both methods are successful on this dataset. However, the MPL approach provides as well a smoothed baseline hazard estimate displayed in Fig. 3. This plot shows that the baseline risk of death, after adjusting for all covariates, peaks after a few days and then reduces gradually until day 28.

We also fitted two proportional hazard models corresponding to albumin and saline treatments, which would be useful in comparing the risks associated with these two treatments if the proportional hazard assumption was violated. Specifically, we fitted these models stratifying the data by treatment arm using MPL in exact the same way as before (i.e. the same smoothing value and bin event count is used). The regression coefficients estimates are summarized in Table 4. The main difference between these two treatments on survival is reflected by a greater risk of swelling and mass lesion for albumin patients. However, it is not clear whether these covariates are important for saline. As the sub-sample is not randomized, caution is nevertheless needed in interpreting these results across treatment arms. Their baseline hazard with 95% pointwise confidence intervals are displayed in Fig. 4. As expected, a much lower baseline risk is observed in the saline arm compared to albumin. The peak at 4 days appearing only the albumin group may suggest a potential deviation from the proportional hazard assumption but caution is needed due to the width of the pointwise confidence intervals.

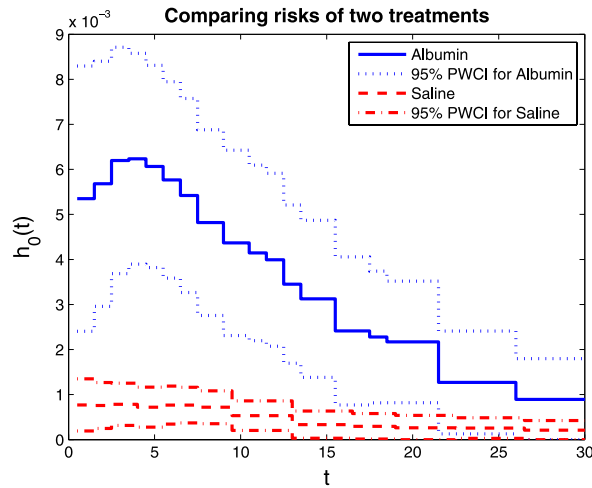


Fig. 4. Plots of baseline hazards and their 95% pointwise confidence intervals (PWCIs) for the two treatments.

6. Conclusions

This paper develops a new algorithm providing simultaneously the MPL estimates of the baseline hazard function and the regression coefficients vector β in proportional hazard models. The algorithm applies alternately the Newton and the MI algorithms in each iteration to update respectively β and θ . This algorithm is very stable and usually converges quickly. $\sqrt{n}$ -consistency and asymptotic normality of MPL is shown for smoothing parameter μ_n of order $o(n^{-1/2})$. Accurate standard errors for the regression coefficients and 95% pointwise confidence intervals around the smoothed estimate of $h_0(t)$ can be easily obtained. Although only one simulation is reported in this paper, all our simulation studies shows that, under independent censoring, MPL provides a small gain in efficiency over PL, especially for small to moderate samples sizes. Under dependent censoring, MPL produces estimates that are less biased than PL leading to a substantial MSE reduction. Whilst the simulations conducted here focus mainly on piecewise approximations to the baseline hazard, other bases such as splines can be used and possibly lead to further efficient gain. One possible advantage of using splines over step functions is that the baseline hazard estimate may avoid artificial leveling-off that can be observed for values of t close to the maximum observed survival time. Although we have developed the method for the proportional hazard model with right censoring, it can easily be extended to other censoring types, such as interval censoring. Such an extension is not straightforward for the PL-based methods. Other survival models like the additive or accelerated failure time models can also be fitted using similar algorithms. We are currently investigating these extensions. Robust loss functions can also be used instead of the log-likelihood. In fact the proof of Theorem 2 is general enough to be used for a large category of problems. Future research also includes efficient ways of determining the optimal smoothing value and the optimal number of bins using general cross validation, particularly for large censoring proportions.

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Appendix A. Proof of Theorem 1

We provide an outline of the proof of Theorem 1.

1. Recall that each iteration of Newton-MI involves two steps: the modified Newton step for updating β where its mapping is denoted by M_1 , and the MI step for updating θ with its mapping denoted by M_2 . M_1 and M_2 satisfy: $\beta^{(k+1)} = M_1(\beta^{(k)}, \theta^{(k)})$ with $\Phi(\beta^{(k+1)}, \theta^{(k)}) \geq \Phi(\beta^{(k)}, \theta^{(k)})$, and $\theta^{(k+1)} = M_2(\beta^{(k+1)}, \theta^{(k)})$ with $\Phi(\beta^{(k+1)}, \theta^{(k+1)}) \geq \Phi(\beta^{(k+1)}, \theta^{(k)})$ and $\theta^{(k+1)} \geq 0$.
2. $X^T A X$ is non-singular if $A^{1/2} X$ has full column rank, and also the gradient given by (14) is finite. Thus the Newton update from Eq. (16) is bounded. We can assume $\beta^{(k)} \in \mathcal{B}$, where $\mathcal{B} = \{\beta : |\beta_j| \leq B < \infty, \text{ for all } j\}$.
3. Update $\theta^{(k+1)}$ from Eqs. (18) and (20) is bounded if $\theta^{(k)}$ is bounded. So we can assume $\theta^{(k)} \in \mathcal{T}$, where $\mathcal{T} = \{0 \leq \theta_u \leq T < \infty, \text{ for all } u\}$.
4. Letting $M = M_2 \circ M_1$, and it defines the iteration mapping for Newton-MI, i.e.

$$(\beta^{(k+1)}, \theta^{(k+1)}) = M(\beta^{(k)}, \theta^{(k)}). \quad (32)$$

Let $\Gamma_\beta = \{\hat{\beta}\}$ be the β stationary points set, i.e. $\partial\Phi(\hat{\beta}, \hat{\theta})/\partial\beta_j = 0$ ($j = 1, \dots, p$), and let $\Gamma_\theta = \{\hat{\theta}\}$ be the θ stationary points set, i.e. $\partial\Phi(\hat{\beta}, \hat{\theta})/\partial\theta_u = 0$ if $\theta_u \neq 0$ and $\partial\Phi(\hat{\beta}, \hat{\theta})/\partial\theta_u < 0$ if $\theta_u = 0$ ($u = 1, \dots, m$). Let $\Gamma = \Gamma_\beta \times \Gamma_\theta$.

For mapping M and the Cartesian product set $\mathcal{B} \times \mathcal{T}$, they satisfy: (i) Mapping M is defined on set $\mathcal{B} \times \mathcal{T}$. (ii) Set $\mathcal{B} \times \mathcal{T}$ is closed and bounded, and thus is compact. (iii) It is easy to verify that M_1 is continuous and hence is closed for all $\beta \in \mathbb{R}^p$, and M_2 is closed for $\theta \notin \Gamma_\theta$. Thus M is close for $(\beta, \theta) \notin \Gamma$. (iv) $\Phi(\beta, \theta)$ is continuous and satisfies $\Phi(\beta^{(k+1)}, \theta^{(k+1)}) \geq \Phi(\beta^{(k)}, \theta^{(k)})$ on $\mathcal{B} \times \mathcal{T}$, where equality holds only when the MPL solution is achieved. According to the Zangwill global convergence theorem (Zangwill, 1969), Newton-MI is convergent if the initials satisfy $|\beta_j^{(0)}| \leq B$ (for all j) and $0 < \theta_u^{(0)} \leq T$ (for all u).

5. Finally, let $\beta^{(k)} \rightarrow \tilde{\beta}$ and $\theta^{(k)} \rightarrow \tilde{\theta}$ as $k \rightarrow \infty$. $\tilde{\beta}$ clearly satisfies the condition that $\partial\Phi(\tilde{\beta}, \tilde{\theta})/\partial\beta = 0$ from Eq. (16). For $\tilde{\theta}$, if $\tilde{\theta}_u \neq 0$, it must have $\partial\Phi(\tilde{\beta}, \tilde{\theta})/\partial\theta_u = 0$ from Eqs. (19) and (20). If $\tilde{\theta}_u = 0$, since $\theta_u^{(0)} > 0$, it must have $N_u = 0$ and $\partial J(\tilde{\beta}, \tilde{\theta})/\partial\theta_u > 0$, thus $\partial\Phi(\tilde{\beta}, \tilde{\theta})/\partial\theta_u < 0$, i.e. $(\tilde{\beta}, \tilde{\theta}) \in \Gamma$.

Appendix B. Proof of Theorem 2

We provide an outline of the proof of Theorem 2.

Firstly, strong consistency can be obtained through the asymptotic results for U -statistics as shown below. For this result we only demand a weaker condition $\mu_n = o(1)$ although $\mu_n = o(n^{-1/2})$ is stated in Condition (4). Note that according to Condition (3) η_0 is an inner point of the compact parameter space. Denote by $\bar{l}(\eta)$ the limit of the averaged likelihood $n^{-1}l(\eta)$.

Let $z_i = (t_i, \xi_i, X_i)$, $i = 1, \dots, n$, be i.i.d. triples. The conjoint density of z_i is

$$g(t_i, \xi_i | x_i; \eta)k(x_i) = g_i(t_i, \xi_i; \eta)k(x_i),$$

where $k(x)$ is the density of the covariate independent of η and $g_i(t_i, \xi_i; \eta)$ is given by (20). Thus the log-likelihood of the triples is $l(\eta) = \sum \log g_i(t_i, \xi_i; \eta)$ up to a constant. The penalized log-likelihood of the triples can be written as

$$\Phi_n(\eta) = n^{-1}\Phi(\eta) = n^{-1}l(\eta) - \mu_n J(\eta).$$

When $\mu_n \rightarrow 0$, $\mu_n J(\eta)$ also tends to zero as $J(\eta)$ is bounded and continuous on the compact parameter space. It follows from the uniform law of large numbers for U -statistics (Honore and Powell, 1994) that $\Phi_n(\eta) \rightarrow \bar{l}(\eta)$ almost surely and uniformly in η in the compact parameter space. This result, together with the uniqueness of $\arg\max_\eta \bar{l}(\eta)$ —see Conditions (3) and (5), implies that $\hat{\eta} \rightarrow \eta_0$ almost surely by application of Corollary 1 of Honore and Powell (1994); see also Cai et al. (2009, Appendix A) for a similar argument in the accelerated failure time model.

The asymptotic normality follows using a standard second order expansion of the objective function. First, note that for n large enough, the MPL estimator $\hat{\eta}$ lies in a neighborhood of η_0 , thus using a Taylor expansion at η_0 yields

$$0 = \frac{\partial\Phi_n(\hat{\eta})}{\partial\eta} = \frac{\partial\Phi_n(\eta_0)}{\partial\eta} + \frac{\partial^2\Phi_n(\tilde{\eta})}{\partial\eta\partial\eta^T}(\hat{\eta} - \eta_0) \quad (33)$$

where $\tilde{\eta}$ is a vector between $\hat{\eta}$ and η_0 . This expansion requires the log-likelihood function to be twice continuously differentiable in this neighborhood as stated in Condition (6). Thus we have

$$\sqrt{n}(\hat{\eta} - \eta_0) = - \left\{ \frac{\partial^2\Phi_n(\tilde{\eta})}{\partial\eta\partial\eta^T} \right\}^{-1} \sqrt{n} \frac{\partial\Phi_n(\eta_0)}{\partial\eta}.$$

Next we need to show that: (a)

$$\sqrt{n} \frac{\partial\Phi_n(\eta_0)}{\partial\eta} \xrightarrow{D} N(0, \mathcal{I}(\eta_0)),$$

and (b)

$$\frac{\partial^2\Phi_n(\tilde{\eta})}{\partial\eta\partial\eta^T} \xrightarrow{P} \mathcal{I}(\eta_0)$$

as $n \rightarrow \infty$. To prove (a), rewrite

$$\sqrt{n} \frac{\partial\Phi_n(\eta_0)}{\partial\eta} = n^{-1/2} \sum_{i=1}^n U(t_i, \xi_i; \eta_0) + \sqrt{n}\mu_n \frac{\partial J(\eta_0)}{\partial\eta}$$

where $U(t_i, \xi_i; \eta_0) = \partial \log g_i(t_i, \xi_i; \eta_0)/\partial\eta = \partial \log g(z_i; \eta)/\partial\eta$, $i = 1, \dots, n$, are the model score functions. The first term on the right hand side involves the sum of n i.i.d. vectors (see Condition (1)). Therefore, by the central limit theorem the first

term converges to $N(0, \mathcal{I}(\eta_0))$, where $\mathcal{I}(\eta_0)$ is the Fisher information matrix given in Condition (6); the second term tends to zero as $\mu_n = o(n^{-1/2})$ and Condition (7), and therefore (a) is proved. For (b) we can rewrite

$$\frac{\partial^2 \Phi_n(\tilde{\eta})}{\partial \eta \partial \eta^T} = n^{-1} \sum_{i=1}^n \frac{\partial^2 \log g(t_i, \xi_i; \tilde{\eta})}{\partial \eta \partial \eta^T} + \mu_n \frac{\partial^2 J(\tilde{\eta})}{\partial \eta \partial \eta^T}.$$

Again the second term of the right hand side tends to 0 as $\mu_n = o(n^{-1/2})$ and Condition (7). Using the consistency property of the MPL $\hat{\eta}$ of η_0 and Condition (6) the first term converges in probability to $\mathcal{I}(\eta_0)$. Finally, the asymptotic normality result follows from (a) and (b); see Amemiya (1983, pp. 333–389).

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